

Course Code- 119405 Structural Analysis**3 1 0 4****Unit- 1.0****8 hrs**

Analysis of indeterminate structures by force methods, flexibility coefficients. unit load method, Influence line and Rolling loads, beam, frames and arches.

Unit- 2.0**5 hrs**

Energy Methods: Principle of minimum potential energy, principle of virtual work, Castigliano's theorems, Reciprocal theorem.

Unit- 3.0**6 hrs**

Muller Breslau Principles and its applications to determinate and indeterminate structures.

Unit- 4.0**8 hrs**

Analysis of Beams and Frames: Moment Area method, Slope deflection method, Three Moment Equation.

Unit- 5.0**5 hrs**

Moments distribution methods, effect of symmetry and ant symmetry, sway correction.

Unit- 6.0**10 hrs**

Lateral load analysis: Portal and Cantilever methods, Matrix method of structural analysis, Displacement/Stiffness methods.

Text/ Reference:-

1. C.S. Reddy, Basic Structural Analysis, Second Edition, Tata McGraw Hill, 2005.
2. R.C. Hibbeler, Structural Analysis, Pearson Education, 6th edition, 2009.
3. C.K. Wang, Intermediate Structural Analysis, Tata McGraw Hill, 1984.